MultiSticker: Intent-Guided Multimodal Sticker Retrieval with Dialogue Memory

Anonymous Author
Department of Computer Science, Rice University
rl182@rice.edu

Abstract

*Stickers are widely used in messaging platforms to express affect, humor and intent. Retrieving the right sticker for a dialogue turn is challenging because (i) the candidate bank is large and dominated by animated formats, (ii) the correct reply depends on long-range conversational state rather than just the last turn, and (iii) each context typically has only one observed positive even though many semantically equivalent stickers exist. We present **MultiSticker**, an intent-guided sticker retrieval system that combines three text streams—recent dialogue context, retrieved long-term session memory, and an LLM-generated reply intent—against a frozen-or-adapted OpenCLIP image bank covering `.png`, `.gif` and `.webm` stickers. We compare a frozen-CLIP zero-shot baseline against a learned retrieval head and three parameter-efficient fine-tuning variants (text-side LoRA, image-side LoRA, dual LoRA) on the U-Sticker corpus (8,129 sessions). On the completed medium-scale run, dual LoRA gives the best trained-model semantic Group@5/10 performance, while image-side LoRA gives the strongest exact top-5 ranking. We argue that for a chat sticker panel, semantic Group@5/10 is the right north-star metric and exact retrieval should be treated as a secondary within-group ranking diagnostic.*

1. Introduction

When users reply with stickers in a chat, the choice depends on local context (what was just said), long-range state (who is talking, prior tone, recurring topics) and the communicative *intent* behind the reply (e.g. gratitude, encouragement, teasing). A retrieval system that only looks at the last few turns frequently selects stickers with the wrong affect or the wrong recurring style for a given user stream. Conversely, a system that uses only image-side similarity ignores the dialogue entirely.

We focus on the U-Sticker corpus, which contains real multilingual chats spanning crypto, gaming and social-domain channels. Several properties make sticker retrieval different from generic image–text retrieval. First, the candidate set is sparse and animated: 42,319 decodable stickers across `.png`, `.gif` and `.webm`. Second, supervision is *single-positive*: each turn records the one sticker the user actually sent, even though many other stickers in the bank are equally valid. Third, turns frequently contain code-switching and chat slang outside CLIP's pre-training distribution.

We build on the IGSR-style design [7] of using a CLIP backbone plus a small learned head, and extend it in three directions:

- A retrieved *long-term memory* stream from previous sessions of the same user/file, encoded by multilingual E5 [8].
- An LLM-generated *reply intent* stream that abstracts away from the exact sticker id and describes the communicative function of the reply.
- A unified pipeline that handles `.png`, `.gif` and `.webm` via per-frame OpenCLIP encoding and mean pooling, and supports parameter-efficient adaptation through LoRA [2] on the text side, image side, or both.

Contributions. (1) A clean five-row experiment matrix (zero-shot CLIP, learned head with memory, text-LoRA, image-LoRA, dual-LoRA) on a unified all-media sticker bank. (2) A medium-scale benchmark showing that dual LoRA gives the most balanced group-level retrieval, while image-side LoRA is strongest for exact top-5 ranking. (3) A reusable, self-contained MultiSticker codebase including data manifest construction, memory retrieval, OpenCLIP wrapping, multi-frame decoding, and SLURM-friendly run scripts.

2. Related Work

Vision–language retrieval. CLIP [6] popularised dual-encoder image–text retrieval, and OpenCLIP [3] provides open-weight reproductions trained on LAION-class corpora. Our system uses ViT-B/32 with the laion2b_s34b_b79k weights as the backbone for both the text and image branches.

Sticker / meme retrieval. Sticker reply suggestion has been studied as a multimodal retrieval and ranking problem [4, 1]. Intent-guided sticker retrieval (IGSR) [10] adds an explicit intent classification head and a group prior. We adopt this template and add long-term memory and parameter-efficient adaptation.

Memory-augmented dialogue. Long-term dialogue memory has been used to condition open-domain chatbots [9]. We use it specifically to construct a third query stream for retrieval rather than to generate text. Our memory encoder is multilingual E5 [8], chosen for code-switched chat data.

Parameter-efficient fine-tuning. LoRA [2] adds low-rank update matrices to selected linear layers, leaving the original weights frozen. We inject LoRA into OpenCLIP's out_proj, c_fc and c_proj modules through PEFT [5], and isolate text-side, image-side and dual adaptation through name-based parameter masking.

3. Method

3.1. Pipeline overview

For each turn t where the user sent a sticker $s_t^\star$, we form three text inputs: the immediate *context* c_t (up to 12 prior turns), the retrieved *memory* m_t (concatenation of the top-3 prior session summaries), and the LLM *intent* i_t (a short description of what the sticker should express). These are encoded independently by the OpenCLIP text encoder f_T. Sticker images s are encoded by the OpenCLIP image encoder f_I (with mean-pooling over frames for animated formats). The trainable head g_θ projects the concatenation into the image space:

$$\mathbf{q}_t = \mathrm{norm}\Big(g_\theta\big([f_T(c_t);\, f_T(m_t);\, f_T(i_t)]\big)\Big). \quad (1)$$

Retrieval logits are $\mathbf{z}_t = \mathbf{q}_t\,\mathbf{B}^\top/\tau$, where $\mathbf{B} \in \mathbb{R}^{N\times d}$ is the (optionally adapted) OpenCLIP image bank and τ is a learned temperature. A second head predicts the sticker's intent group y_t, and we minimise

$$\mathcal{L} = \mathrm{CE}(\mathbf{z}_t, s_t^\star) + \lambda\,\mathrm{CE}(\mathbf{z}_t^{\mathrm{int}}, y_t), \quad \lambda{=}0.2. \quad (2)$$

3.2. Long-term memory retrieval

Each raw chat file is split into sessions by a 12-hour inactivity gap. For sample t, we collect up to 8 prior sessions from the same file. Their LLM-generated session summaries are encoded with the E5 passage: prefix; the current session text is encoded with the query: prefix. After L2 normalisation, the dot products rank the prior sessions and we keep the top-3. The selected memory texts are concatenated and stored as m_t, then encoded by OpenCLIP at training time. Crucially, memory enters the model through the *text* encoder, not as a separate vector, which lets the retrieval head learn to reweight memory against the local context.

3.3. LLM annotation

We annotate every session with a $\leq$ 4-sentence session_memory_text (used as the passage corpus for memory retrieval), and every sticker-reply sample with one intent_label from a 10-class taxonomy (celebration_approval, humor_teasing, encouragement_support, gratitude, affection_care, confusion_curiosity, frustration_disapproval, disappointment_sadness, surprise_reaction, neutral_acknowledgment) plus a free-form intent_text. The intent label feeds the auxiliary head and induces the semantic-group label space; the intent text becomes the third query stream.

3.4. Multi-frame sticker encoding

.png stickers are encoded as one-frame clips. .gif and .webm are decoded to all frames (FFmpeg for video, PIL for GIF), each frame is encoded by the OpenCLIP image encoder, and the per-frame embeddings are mean-pooled and renormalised. During image-LoRA training, the full bank is re-encoded once per epoch without gradients; for each batch we additionally re-encode only the unique *positive* stickers with gradients (animated positives capped at 4 frames) and patch them into a cloned bank, which keeps backprop tractable for a 42K-sticker bank.

3.5. Tuning modes

We compare four tuning modes against a zero-shot CLIP baseline. **direct_clip** uses no learned head: the query is just $f_T(c_t)$ scored against the frozen image bank. **head_only** (*memory*) trains the retrieval head on top of frozen OpenCLIP encoders. **text_lora** additionally injects LoRA (r=8, α=16) into the text branch. **image_lora** injects LoRA into the visual branch only, while **dual_lora** adapts both branches. Only LoRA parameters and the head are trainable; all original OpenCLIP weights are frozen.

4. Experiments

4.1. Setup

We report the completed rebuild_medium setting. The manifest is constructed from 8,129 sessions and 368,503 provisional sticker-reply samples. After frequency filtering and media decoding, the experiment uses 30,000 training samples, 5,000 validation samples (4,998 after media filtering), and a 6,758-sticker bank over .png, .gif, and .webm. All trainable models are optimized for 5 epochs with AdamW. The retrieval head uses batch size 256; LoRA runs use batch size 96; inference batch size is 256. The CLIP backbone is OpenCLIP ViT-B/32 with

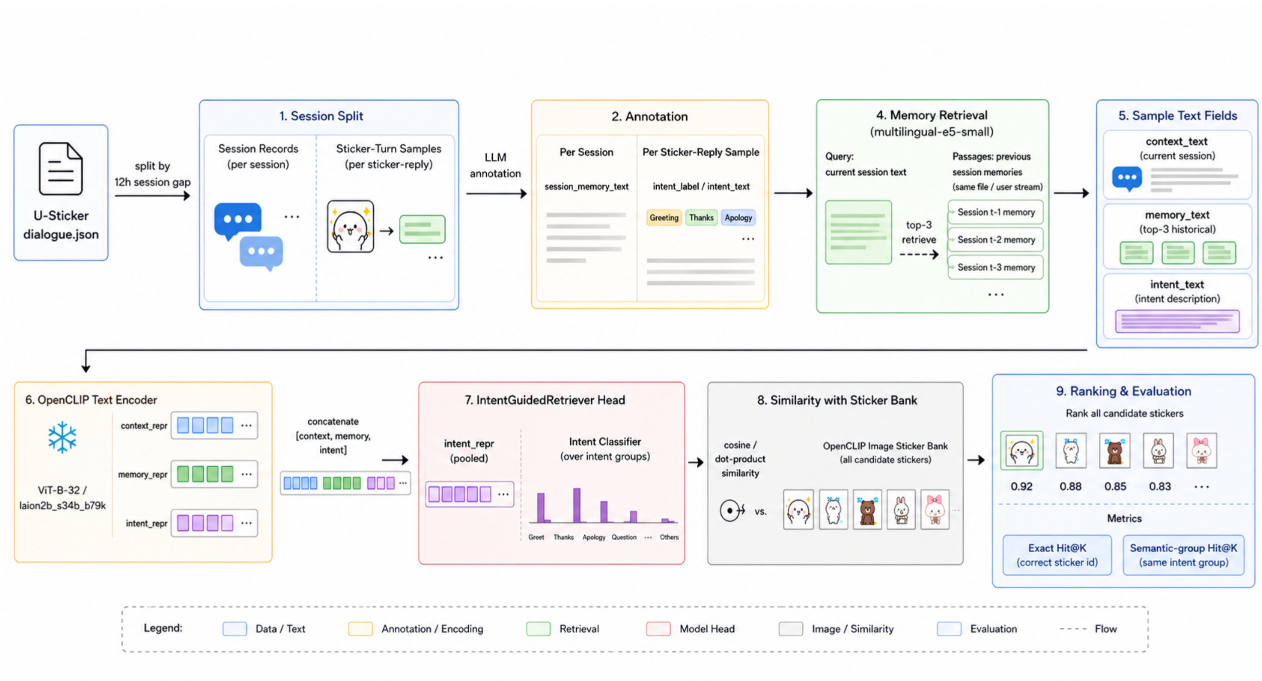

Figure 1. End-to-end MultiSticker pipeline. Raw U-Sticker JSON is split into 12-hour sessions; an LLM produces a per-session memory and a per-sample intent label/text; multilingual E5 retrieves the top-3 historical sessions for the current turn; OpenCLIP independently encodes the context, memory and intent strings; the `IntentGuidedRetriever` head projects their concatenation into the CLIP image space and ranks the sticker bank by cosine similarity. An auxiliary intent classifier predicts the sticker's group and supplies a prior for the two-stage rerank.

`laion2b_s34b_b79k` weights. The retrieval temperature is 0.07, hidden dimension is 512, dropout is 0.1, the auxiliary intent-loss weight is 0.2, and the intent vocabulary is clustered into 64 groups. LoRA uses rank r=8, α=16, dropout 0.05, LoRA LR 10^{-4}, and head LR 10^{-3}.

Compared models. We evaluate five rows. `direct_clip` is a zero-shot baseline with no learned parameters. `head_only` freezes OpenCLIP and trains only the retrieval and intent heads. `text_lora`, `image_lora`, and `dual_lora` train the head plus LoRA adapters on the text branch, the visual branch, or both branches respectively. This design isolates whether domain adaptation is more useful on chat text, sticker imagery, or the shared alignment geometry.

4.2. Metrics

We evaluate retrieval at two granularities. The primary metric is **Group Hit@K**: a prediction is correct if any of the top-K retrieved stickers belongs to the same semantic intent group as the logged sticker. This is the most faithful proxy for a chat sticker panel, where the interface shows 5–10 plausible choices and the user mainly needs the right affect, tone, or communicative intent. We therefore treat **Group@5** and **Group@10** as the north-star metrics. As a secondary diagnostic, we also report **Exact Hit@K**, which requires recovering the single observed sticker id. Exact metrics measure within-group ranking quality, but are conservative because many unobserved stickers may be equally valid. All Hit@K metrics are higher-is-better.

4.3. Results

Table 1 and Table 2 summarise the completed medium-scale run. The key result is that `dual_lora` is the best trained model under the panel-recommendation objective: it wins overall Group@5 and is second only to the zero-shot baseline on Group@10. At the same time, `image_lora` has the strongest exact top-5 ranking, which suggests that visual adaptation is especially useful for ordering near-duplicate stickers within a semantic group.

5. Analysis and Discussion

Group retrieval is the right north star. The most important deployment regime is not auto-sending a single sticker; it is filling a chat panel with 5–10 plausible choices. In this regime, recovering the *semantic group* of the desired reply is more important than recovering the exact historical sticker id. The results support this framing: exact retrieval remains difficult for all methods, but Group@5 and Group@10 are already near the level re-

Model	Trainable	Group@1↑	Group@5↑	Group@10↑	Group@30↑	Exact@5↑	Exact@30↑	MAP↑
direct_clip	none	0.6335	0.8778	**0.9084**	**0.9588**	0.0012	0.0028	0.0013
head_only	head	**0.7933**	0.8619	0.8780	0.9406	0.0296	**0.1263**	0.0282
text_lora	head+T	**0.7623**	0.8661	0.8880	0.9420	0.0312	0.1190	0.0280
image_lora	head+V	0.7063	**0.8812**	0.9044	**0.9550**	**0.0376**	0.1188	**0.0302**
dual_lora	head+T+V	0.6275	**0.8816**	**0.9064**	0.9540	**0.0326**	**0.1315**	**0.0286**

Table 1. **Main rebuild_medium results.** The north-star metrics are Group@5 and Group@10, matching a chat UI that displays a small panel of sticker candidates. Exact metrics are stricter single-positive diagnostics. **Red bold** marks the best method per column and **blue bold** marks the runner-up. T/V denote text/visual LoRA.

Media	Model	Count	Group@5↑	Group@10↑	Exact@5↑	Exact@30↑
PNG	head_only	1,676	0.9141	0.9242	0.0197	**0.1062**
PNG	text_lora	1,676	0.9206	0.9314	**0.0286**	**0.0990**
PNG	image_lora	1,676	**0.9254**	**0.9391**	0.0185	0.0752
PNG	dual_lora	1,676	**0.9320**	**0.9403**	**0.0304**	0.0973
GIF	head_only	2,284	0.8695	0.8870	**0.0499**	0.1935
GIF	text_lora	2,284	0.8717	0.8971	0.0468	0.1843
GIF	image_lora	2,284	**0.8927**	**0.9168**	**0.0670**	**0.1948**
GIF	dual_lora	2,284	**0.8892**	**0.9177**	0.0464	**0.2062**
WEBM	head_only	1,038	0.7611	0.7832	0.0010	0.0106
WEBM	text_lora	1,038	0.7659	0.7977	0.0010	0.0077
WEBM	image_lora	1,038	**0.7842**	**0.8208**	**0.0039**	**0.0222**
WEBM	dual_lora	1,038	**0.7832**	**0.8266**	**0.0058**	**0.0222**

Table 2. **Per-media breakdown for trainable models.** Direct CLIP is omitted because its result artifact does not store per-media metrics. The group metrics are strongest and most stable in PNG/GIF, while WEBM remains the hardest exact-retrieval format.

quired for a usable recommendation panel. Under this criterion, dual_lora is the best trained model: it achieves the strongest Group@5 (0.8816) and the strongest trained-model Group@10 (0.9064). The zero-shot direct_clip baseline slightly exceeds it on Group@10, but its exact ranking is nearly unusable, indicating that CLIP recognises coarse semantic neighborhoods while failing to distinguish the concrete sticker vocabulary.

Exact metrics are secondary but still diagnostic. Exact Hit@K should not be treated as the primary objective because U-Sticker contains only one logged positive per context. A different sticker with the same intent may be a perfectly good recommendation but is counted as wrong by Exact Hit@K. Still, exact metrics reveal how well a model orders siblings inside a group. Here image_lora has the best Exact@5 and MAP, suggesting that visual adaptation improves fine-grained ordering among similar stickers. dual_lora, meanwhile, has the best Exact@30, which is attractive for a two-stage system: first retrieve a semantically correct candidate pool, then rerank or diversify it for display.

Why LoRA helps different media differently. The per-media table shows that GIF is the most responsive format to visual adaptation: image_lora wins GIF Exact@5 and Group@5, while dual_lora wins GIF Exact@30 and Group@10. This is intuitive: animated stickers encode much of their intent in character pose, motion, and style, so adapting the visual branch makes the embedding space more sticker-specific. PNG behaves differently. The frozen CLIP space already separates many static stickers well, so head_only remains strong on PNG Exact@30, while dual_lora gives the best PNG Group@5/10. WEBM remains the hardest exact-retrieval case; frame sampling and video compression likely erase cues needed to distinguish close variants.

Recommended retrieval design. For a production-style sticker panel, the cleanest design is to use dual_lora as the group-aware first-stage retriever and optimize the system around Group@5/10. Exact@5 and MAP should be tracked as secondary ranking signals, where image_lora currently sets the strongest reference point. A future reranker can combine these strengths: use dual_lora to guarantee semantic coverage, then apply a lightweight visual/textual reranker or diversity rule to choose the final 5–10 stickers.

6. Conclusion

We presented MultiSticker, an intent-guided dialogue-aware sticker retrieval system that combines OpenCLIP, retrieved long-term memory, and LLM-generated intent into a single trainable head with optional LoRA adaptation on either encoder branch. On the completed medium-scale U-Sticker run, dual LoRA is the strongest trained model for group-level panel recommendation, while image-side LoRA gives the best exact top-5 ordering. This supports a two-stage view of sticker retrieval: optimize the first stage for semantically correct 5–10 item panels, then use exact metrics as secondary diagnostics for fine-grained reranking. The released code makes the five-row comparison reproducible end-to-end.

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